01.

North South University

Department of Electrical & Computer Engineering

LAB REPORT-01

Course Code: EEE111L

Course Title: Analog Electronics

20/20

Section: 05

Lab Number: 01

Experiment Name:

I-V Characteristics of Diode

Experiment Date: 16/02/2023

Date of Submission:09/03/2023

Group members:

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Course Instructor: Mohammad Shorif Uddin (MHUD)

Submitted To Lab instructor : Md. Masud Rana

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2. Experiment Name : I-V Characteristics of diode.

3. Objective : To study the I-V characteristics of diode.

4. Theory :

A diode is a two-terminal electronic component that conducts current primarily in one direction (asymmetric conductance); it has low (ideally zero) resistance in one direction, and high (ideally infinite) resistance in the other. It is a bi-polar device that behaves as the short circuit when it is in ~~forward~~ bias and as an open circuit when it is in reverse bias condition.

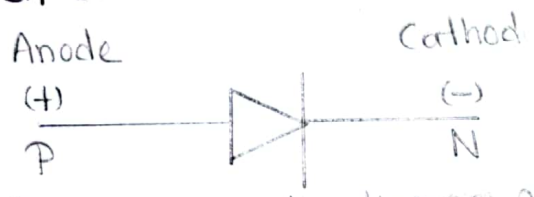


Figure: Schematic diagram of Diode.

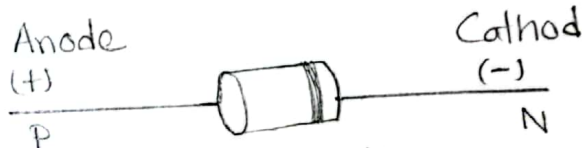


Figure: P-N junction Diode

The most common type of diode uses a p-n junction. In this type of diode, one material (n) in which electrons are charge carriers and a second material (p) in which holes (places depleted of electrons that act as positively charged particles) act as charge carriers.

There are two types of biasing condition for a diode:

1. When the diode is connected across a voltage

source with positive polarity of source connected to p side of diode and negative polarity to n side, then the diode is in forward bias condition.

2. When the diode is connected across a voltage source with positive polarity of source connected to n side of diode and negative polarity to p side, then the diode is in reverse bias condition.

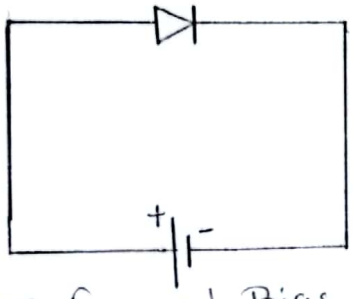


Figure: Forward Bias Condition

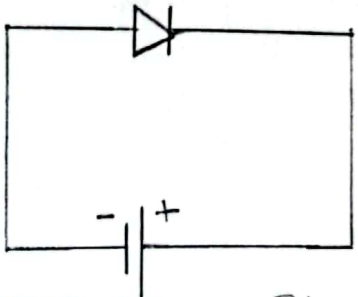


Figure: Reverse Bias Condition

If the input voltage is varied and the current through the diode corresponding to each voltage are taken, the plot of diode current (I_D) Vs diode voltage (V_D) will be follows;

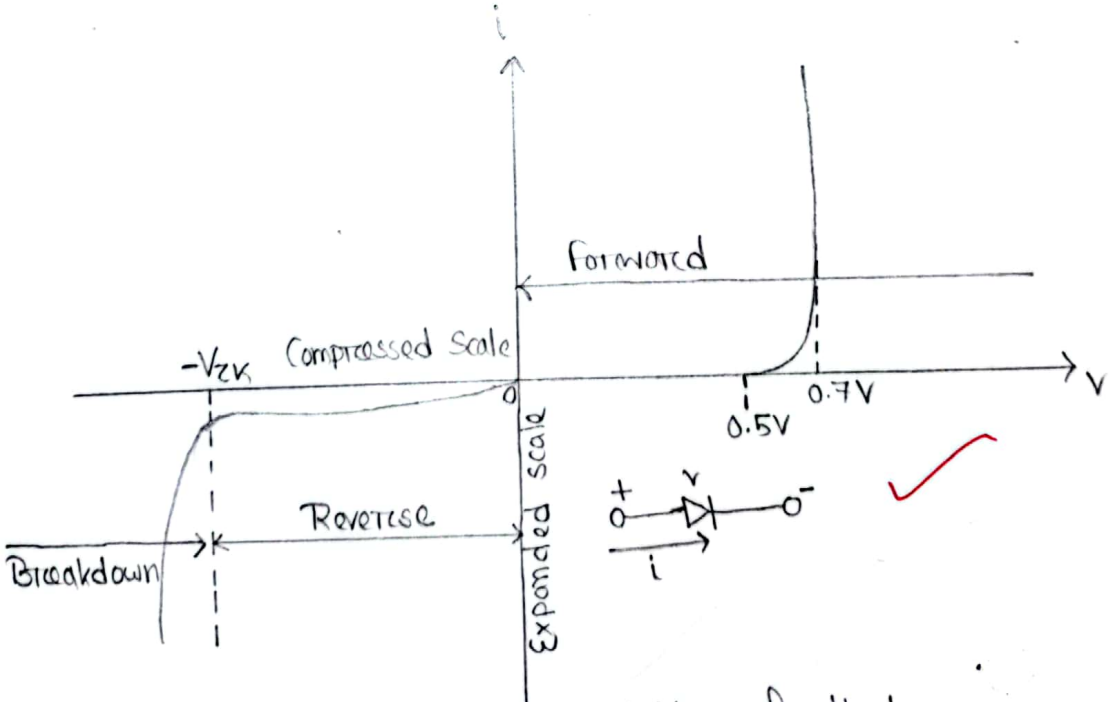


Figure: I-V characteristics of diode

At the reverse bias condition the amount of current flows through the diode is very small but if the voltage continuously increases in reverse direction, at a certain value that diode will break down and huge amount of current will flow in reverse direction. This is called breakdown of diode.

From the characteristics curve it can be seen that, a particular forward bias voltage (V_T) is required to reach the region of upward swing. This voltage V_T is called the cut-in voltage or threshold voltage of diode. For Si diode the typical value of threshold voltage is 0.7 volt and for Ge diode is 0.3 volt.

5. Apparatus:

Serial No.	Component Details	Specification	Quantity
01.	p-n junction diode	1N4007	1 piece
02.	Resistor	1 kΩ	1 piece
03.	DC power supply		1 unit
04.	Digital Multimeter		1 unit
05.	Chords and Wire		as required

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6. Circuit Diagram :

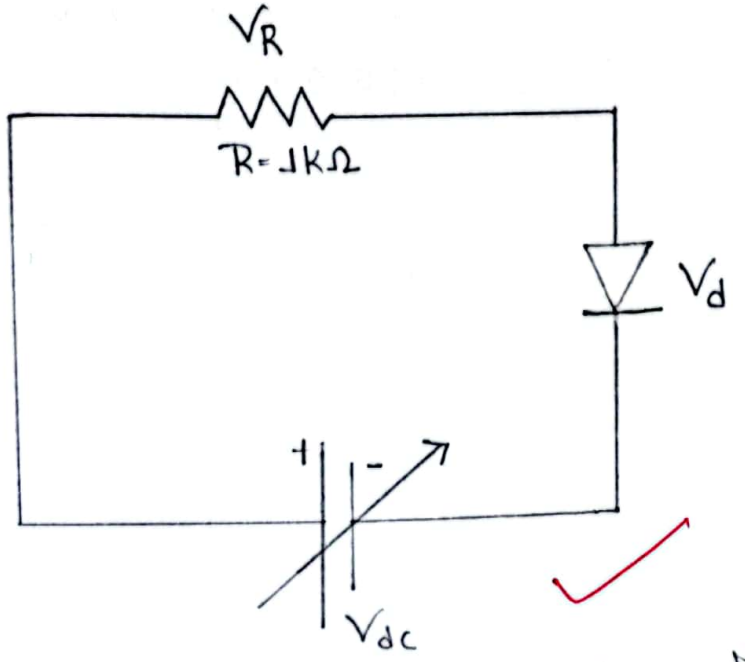


Figure: Diode - forward biased configuration.

7. Experimental procedure:

i) Firstly, we need to measure the resistance (1KΩ) accurately using DMM.

ii) Next, we have to construct the circuit and vary input voltage V_{dc} with values given in table.

iii) Then, we need to measure V_d and V_R for each increment of V_{dc} and record data.

iv) Lastly, we have to calculate the values of I_d using the formula, $I_d = V_R / R$.

8. Results :

Theoretical value : R = 1kΩ

Measured value : R = 0.987 kΩ

Theoretical value:

V_{dc} (Volt)	V_d (volt)	V_R (volt)	$I_d = V_R/R$ (mA)
0.1	0.099V	0V	0mA
0.3	0.291V	0.008V	0.008mA
0.5	0.410V	0.089V	0.089mA
0.7	0.461V	0.238V	0.238mA
0.9	0.489V	0.410V	0.410mA
1	0.499V	0.500V	0.500mA
2	0.554V	1.445V	1.445mA
3	0.581V	2.419V	2.419mA
4	0.598V	3.401V	3.401mA
6	0.622V	5.377V	5.377mA
8	0.638V	7.361V	7.361 mA
10	0.651V	9.349V	9.349 mA
12	0.661V	11.339V	11.339mA
14	0.669V	13.330V	13.330mA
16	0.677V	15.323V	15.323mA
18	0.683V	17.316V	17.316mA
20	0.689V	19.311V	19.311mA

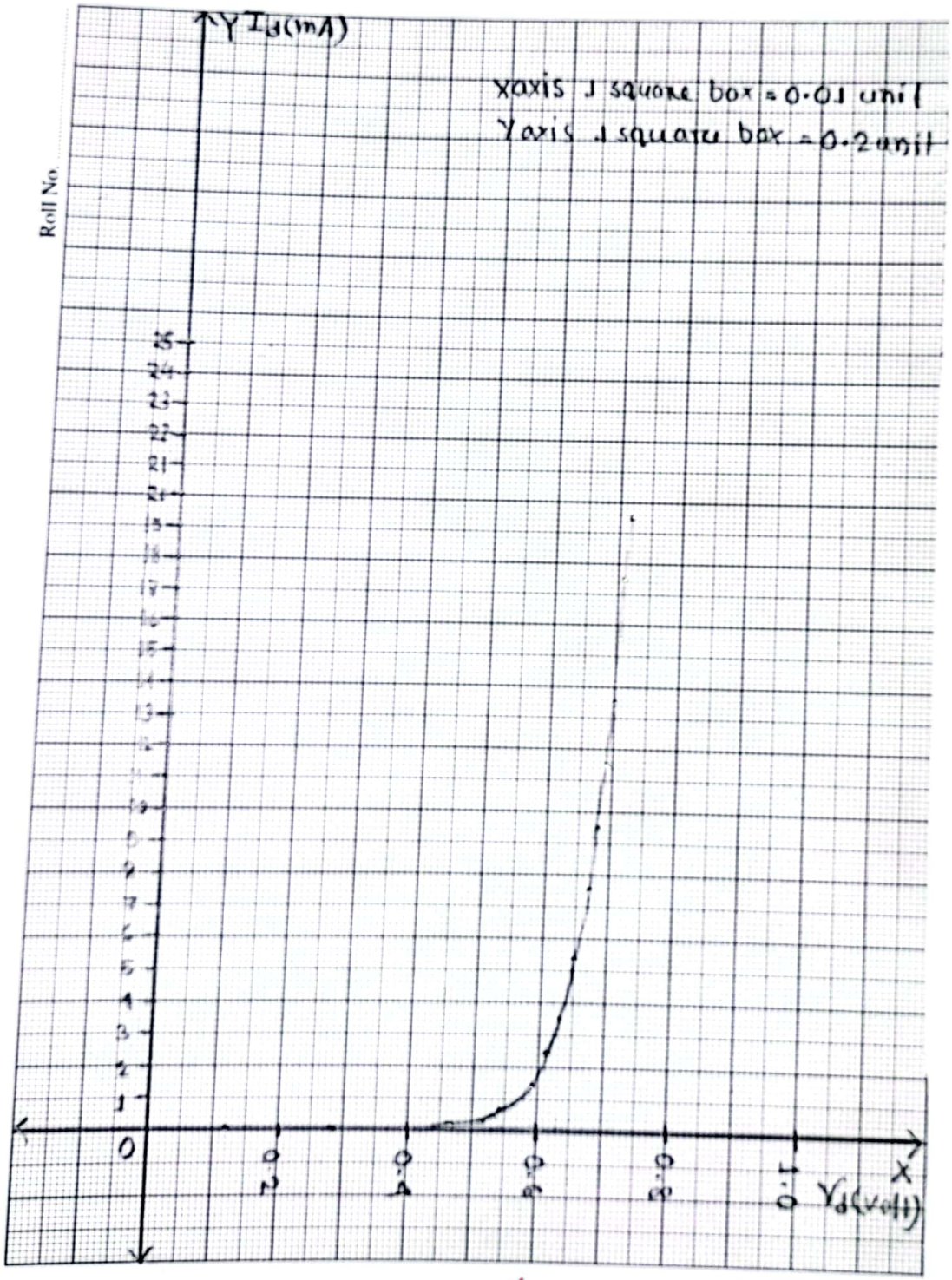
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Measured Value:

V_{dc} (volt)	V_d (volt)	V_R (volt)	$I_d = V_R/R$ (mA)
0.1	0.129V	0V	0mA
0.3	0.330V	0V	0 mA
0.5	0.478V	0.11V	0.11mA
0.7	0.511V	0.24V	0.243mA
0.9	0.534V	0.40V	0.41 mA
1	0.540V	0.525V	0.53mA
2	0.593V	1.467V	1.48mA
3	0.618V	2.462V	2.49 mA
4	0.635V	3.459V	3.50 mA
6	0.656V	5.37V	5.44mA
8	0.671V	7.43V	7.52mA
10	0.682V	9.34V	9.46mA
12	0.691V	11.36V	11.51 mA
14	0.699V	13.38V	13.55mA
16	0.705V	15.30V	15.50mA
18	0.710V	17.31V	17.54 mA
20	0.715V	19.33V	19.58 mA

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Graph:



Graph Name?

10. Discussion :

In the laboratory, we carried out an experiment to study the I-V characteristic of diode. In this experiment we have learned about the forward bias condition of the diode. Firstly, we measured the resistance of resistor using a multimeter. The theoretical value of the resistance is 1kΩ, but our measured value is 0.987 kΩ. We constructed the circuit on the breadboard. We varied the input voltage from 0.1V to 20V. We measured V_d and V_R by taking readings across the circuit diode and resistor respectively using the multimeter. We then calculated I_d using the formula $I_d = V_R/R$.

While conducting the experiment, we have observed some fluctuation between the theoretical and practical value. The reason behind these errors can be a malfunction of a hardware component or unregulated power supply of any sort. The theoretical value of resistor is higher than the measured value because real resistors always have tolerance that make them more or less than their marked theoretical schematic values. In our experiment when we have given the input voltage of 8 V we have measured the $I_d = 7.52$ mA. But by calculating theoretically (from multisim) we have got $I_d = 7.361$ mA.

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$$\% \text{ error} = \left| \frac{\text{Measured Value} - \text{Theoretical Value}}{\text{Theoretical Value}} \right| \times 100\%$$

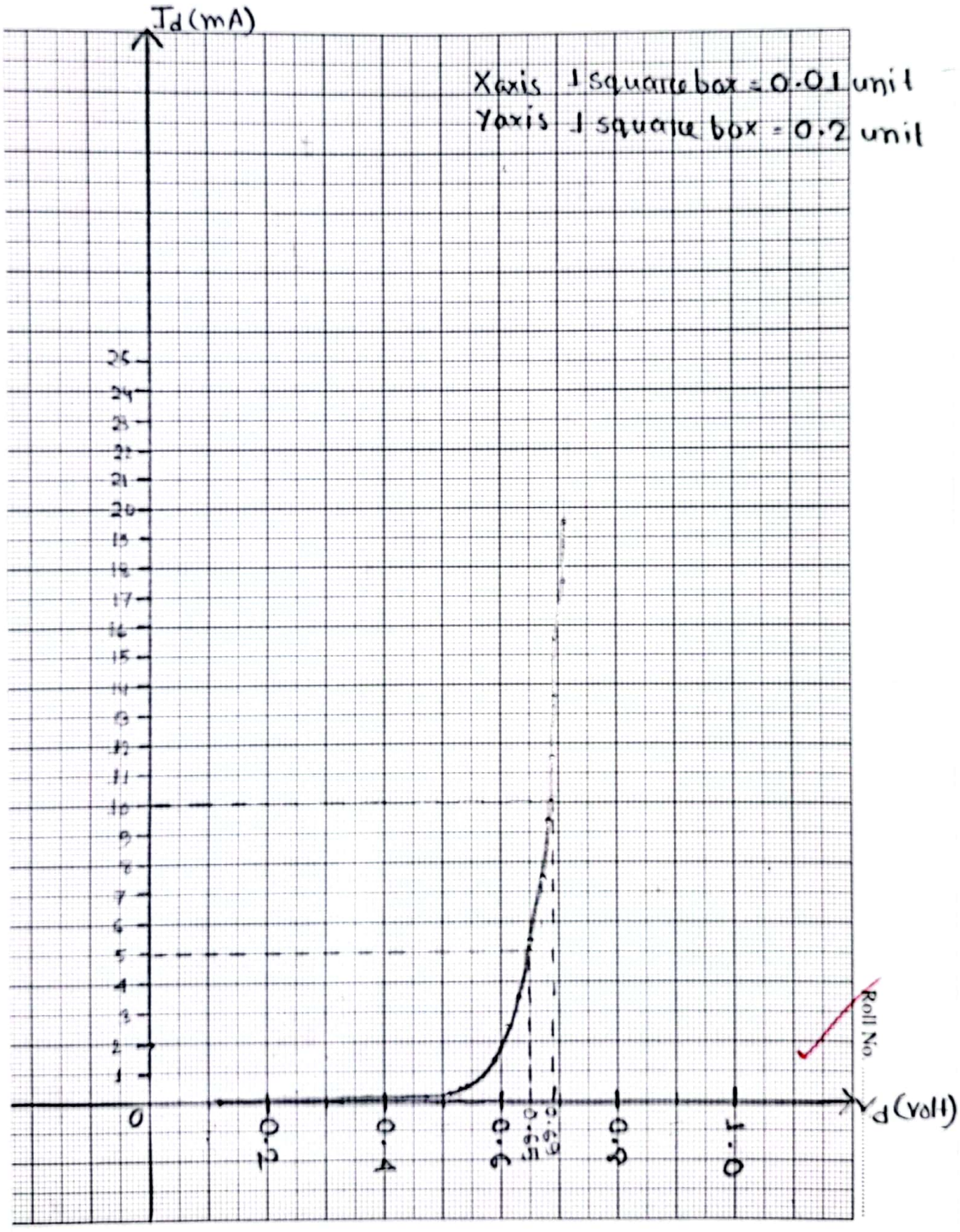
$$= \left| \frac{7.52\,mA - 7.361\,mA}{7.361\,mA} \right| \times 100\%$$

$$= 2.16\%$$

Therefore, we have got the percentage of error of 2.16 % which is caused because of environmental error, procedural error, human error, instrumental error etc. But while doing the experiment I have not faced any big issue as we did it in offline lab and our lab instructor has helped us in every possible way.

9. Questions and Answers:

01. Solution:



02. Solution:

Dynamic resistance or AC resistance:

The dynamic resistance is the resistance offered by the p-n junction diode when AC voltage is applied. When forward bias voltage is applied to a diode that is connected to AC circuit, an AC or alternating current flows through the diode. In AC circuit, charge carriers or electric current does not flow in single direction. It flows in both forward and reverse direction. Dynamic resistance is also defined as the ratio of change in voltage to the change in current. It is denoted as r_f,

$$r_f = \frac{\text{change in voltage}}{\text{change in current}}$$

Static resistance or DC resistance:

When forward bias voltage is applied to a diode that is connected to a DC circuit, a DC or direct current flows through the diode. Direct current or electric current is nothing but the flow of charge carriers (free electrons or holes) through a conductor. In DC circuit, the charge carrier flow steadily in single direction or forward direction. The resistance offered by a p-n junction diode when it is connected to a DC circuit is called static resistance. Static resistance is also defined as the ratio of DC voltage applied across diode

to the DC current or direct current flowing through the diode. The resistance offered by the p-n junction diode under forward biased condition is denoted as R_f.

$$R_f = \frac{\text{DC voltage}}{\text{DC current}}$$

Values.

03. Solution:

From the graph (drawn in question No: 01);

Where, $I_d = 5mA$ $V_d = 0.65\,V$

Where, $I_d = 10mA$ $V_d = 0.69V$

∴ For $I_d = 5mA$ and $V_d = 0.65V$

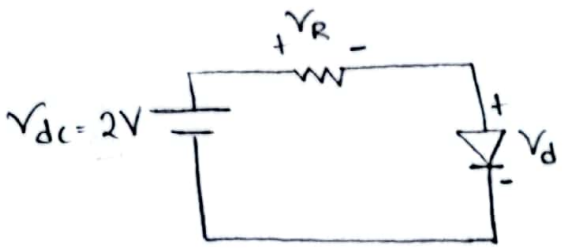
static resistance, $R_f = \frac{V_d}{I_d} = \frac{0.65\,V}{5mA} = 0.13K\Omega$

∴ For $I_d = 10mA$ and $V_d = 0.69V$

static resistance, $R_f = \frac{0.69V}{10mA} = 0.069K\Omega$

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04. Solution:



$I_d = I_R = I$

Applying KVL,

$$V_{dc} - V_d - V_R = 0$$

$$\Rightarrow 2 - V_d - IR = 0$$

$$\Rightarrow 2 - IR - 0 = 0 \quad [V_d = 0]$$

$$\Rightarrow 2 = IR$$

$$\Rightarrow I = \frac{2}{R} = \frac{2}{1k\Omega} = 2mA$$

When, $I = 0$

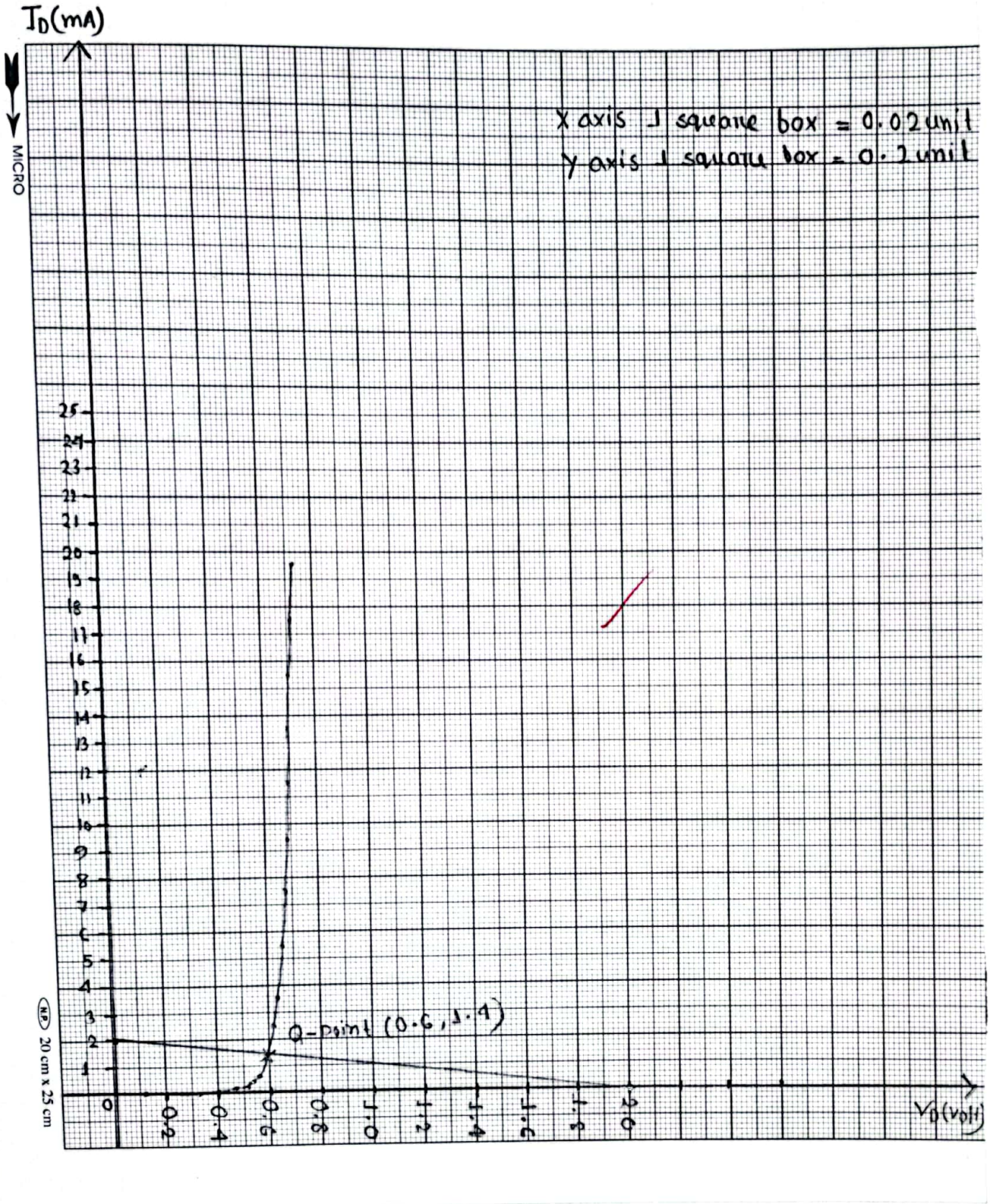
$$\Rightarrow 2 - 0 - V_d = 0$$

$$\therefore V_d = 2V$$

$\therefore$ Load line are (0,2) and (2,0).

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5. Solution:



from the graph we can see that the load line cuts the curve at Q(0.6, 1.15)

NORTH SOUTH UNIVERSITY

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Data Collection: *Signature of instructor:*

Experiment: 1
Performed by Group#

Theoretical value: R = 1 kΩ

Measured value: R = 0.987 kΩ 0.987 KΩ

Masud
23/023/2023

V_{dc} (volt)	V_d (volt)	V_R (volt)	$I_d = V_R / R$ (mA)
0.1	0.129V	0V	0 mA
0.3	0.33V	0V	0 mA
0.5	0.478V	0.11V	0.11 mA
0.7	0.511V	0.24V	0.243 mA
0.9	0.534V	0.40V	0.41 mA
1	0.540V	0.525V	0.53mA
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14	0.699V	13.38V	13.55 mA
16	0.705V	15.30V	15.50 mA
18	0.710V	17.91V	17.54 mA
20	0.715V	19.33V	19.58 mA

experiment Values.

Report:

1. Taking readings from the data table, draw I_d VS V_d curve of diode in a graph paper with proper scale [x-axis: 0.2 V per unit, y-axis: any suitable range].
2. What is dynamic and static resistance of a diode?
3. From the graph, find Vd for corresponding values of I_d = 5 mA and I_d = 10 mA and calculate the static resistance.
4. Considering V_{dc} = 2 volt, find the load line (Showing all calculations)
5. Draw the load line in the I_d VS V_d curve of diode and find Q-point.

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Experimental Setup:

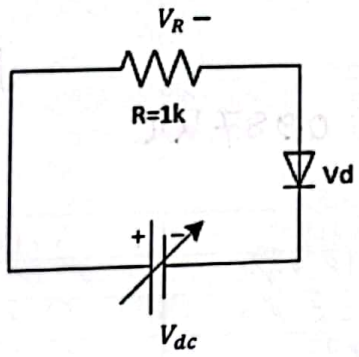
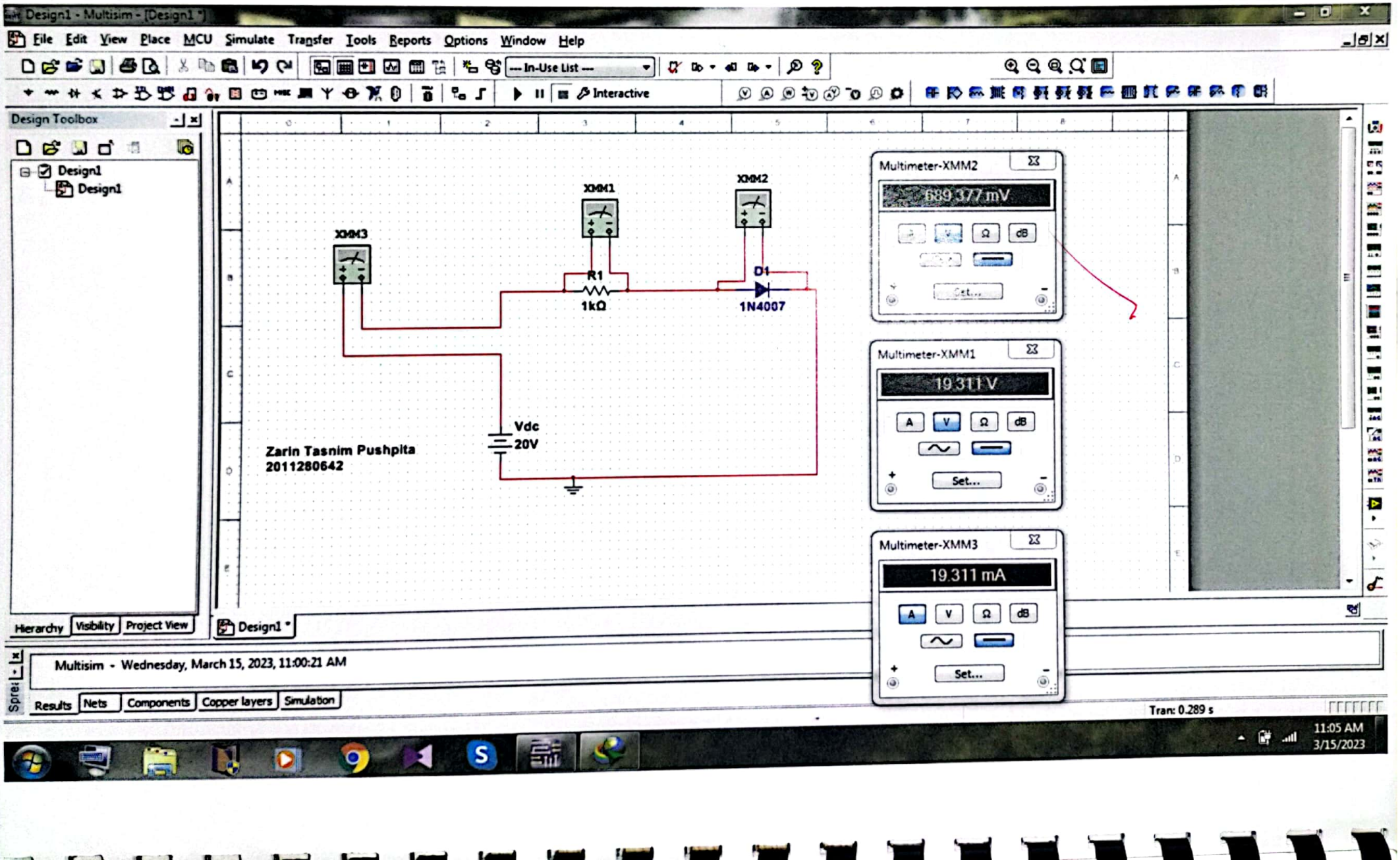


Figure 1.6: Diode forward biased configuration

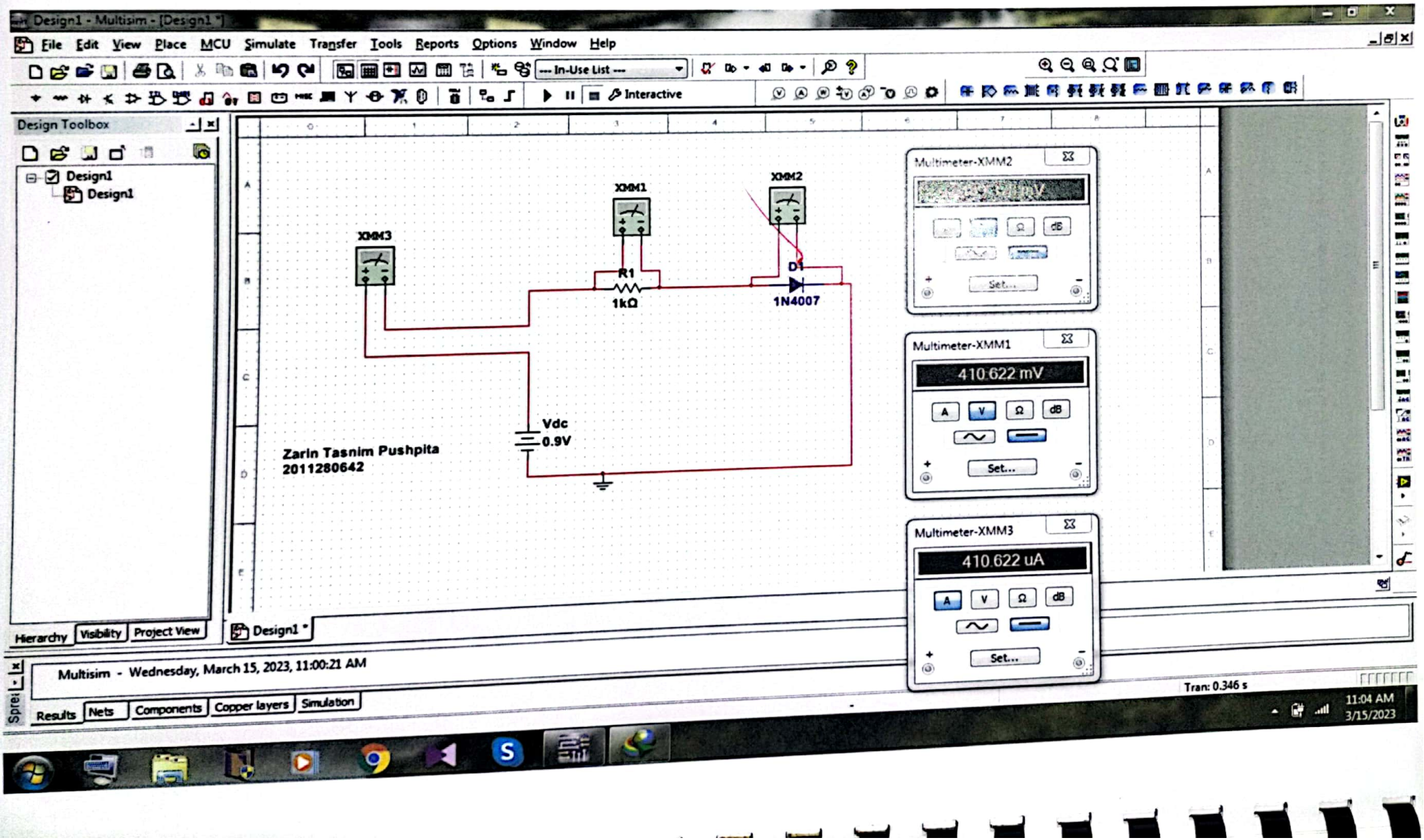
Procedure:

1. Measure the resistance accurately using DMM.
2. Construct the circuit as shown in Figure 1.6.
3. Vary input voltage V_{dc} with values given in table.
4. Measure V_{dc} , V_d and V_R for each increment of V_{dc} and record data on data table.
5. Obtain maximum value of V_d without exceeding 25V for V_{dc}
6. Calculate the values of I_d using the formula, $I_d = V_R / R$.

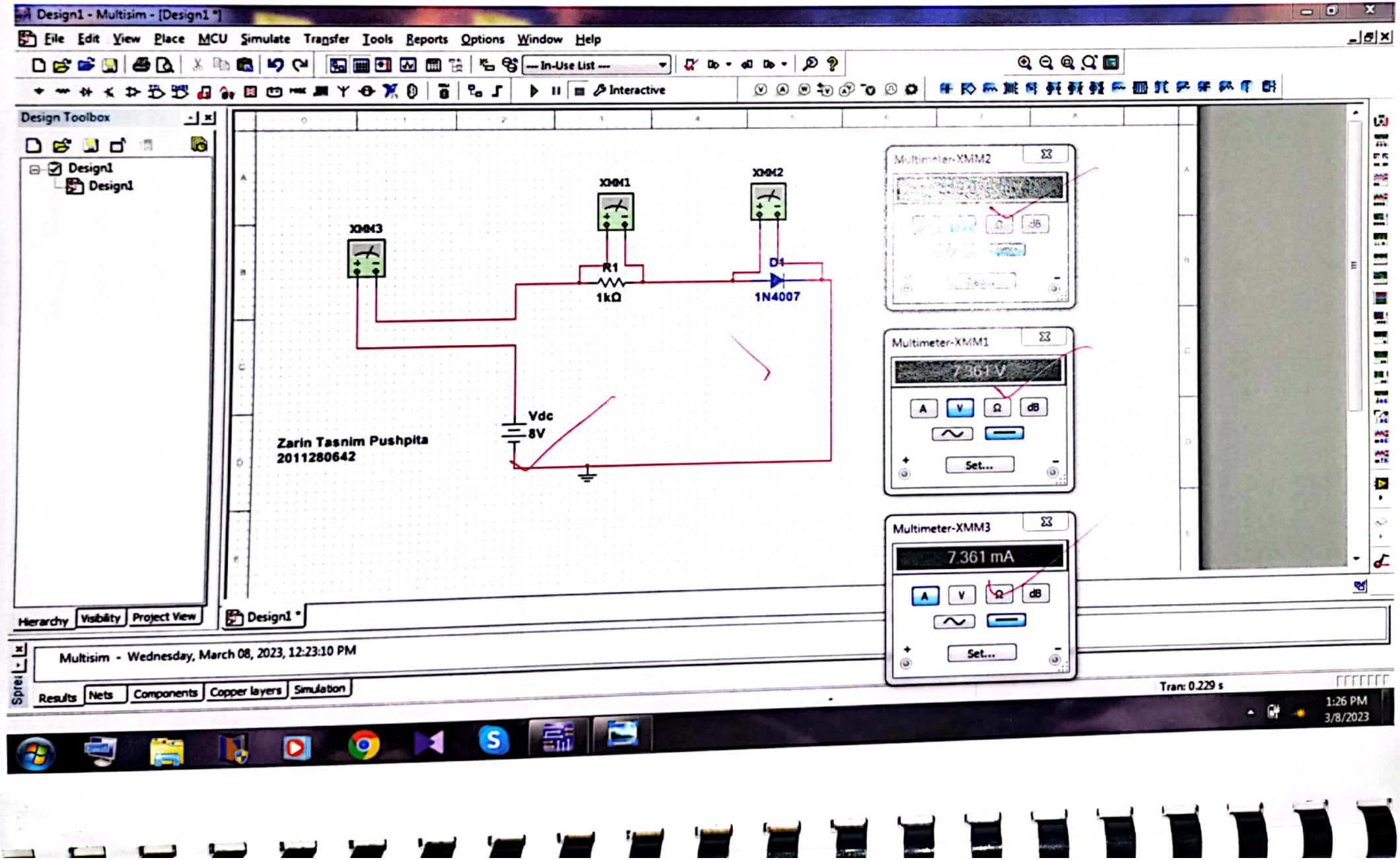
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Design1
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XMM2
XMM3
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1kΩ
D1
1N4007
Vdc
20V
Zarin Tasnim Pushpita
2011280642
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Multimeter-XMM1
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Multimeter-XMM3
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Rania Noor
2011492642

Discussion: In this experiment we could observe that the theoritical value and the measured value slightly differ from each others.

For the experimental value ~~Only the first vo~~ the first two values of V_D and I_d are 0. But the for 0.1V, the measured value is zero. Also, I_d of measured value ~~is~~ slightly ~~flactuates~~ the ~~the~~ V_R. Whereas, in theoritical values the V_d and I_d is always the same. The reason behind the ~~flactuation~~ for measured value is that the resistor is not exactly 1kΩ ~~(R=0.987k)~~ (R=0.987kΩ). As we know, $I_d = V_R/R$. For that reason, V_R and I_d vary a little bit. On the ~~Ot~~ other hand, for the theoritical values the Resistor ~~(R=1kΩ)~~ is always ~~constan~~ 1kΩ. Due to that, the V_R and I_d values are always unchanged. However, in both cases the V_R and I_d values are close to the V_{dc}. ~~for~~ ~~Good~~ Subconsequently, the value of V_d is a

V_{dc} (volt)	V_d (volt)	V_R (volt)	$I_d = V_R/R$ (mA)
0.1	0.099V	0	0
0.3	0.291 V	0.008V	0.008mA
0.5	0.410 V	0.089 V	0.089mA
0.7	0.461 V	0.238V	0.238 mA
0.9	0.489V	0.410 V	0.410mA
1	0.499	0.500	0.500 mA
2	0.554V	1.445 V	1.445 mA
3	0.581 V	2.419V	2.419mA
4	0.598 V	3.401 V	3.401mA
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14	0.669V	13.330V	13.330mA
16	0.677V	15.323V	15.323mA
18	0.683V	17.316 V	17.316 mA
20	0.689 V	19.311 V	19.311 mA

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little bit high # for the measured values than the theoritical values. Also, ~~the Id is great~~ Id values are negligibly greater for measured values than the theoritical values.

Since the theoritical values and measured ~~m~~ value do not ~~match~~ match, it is evident that ~~is a~~ there are errors in this experiment.

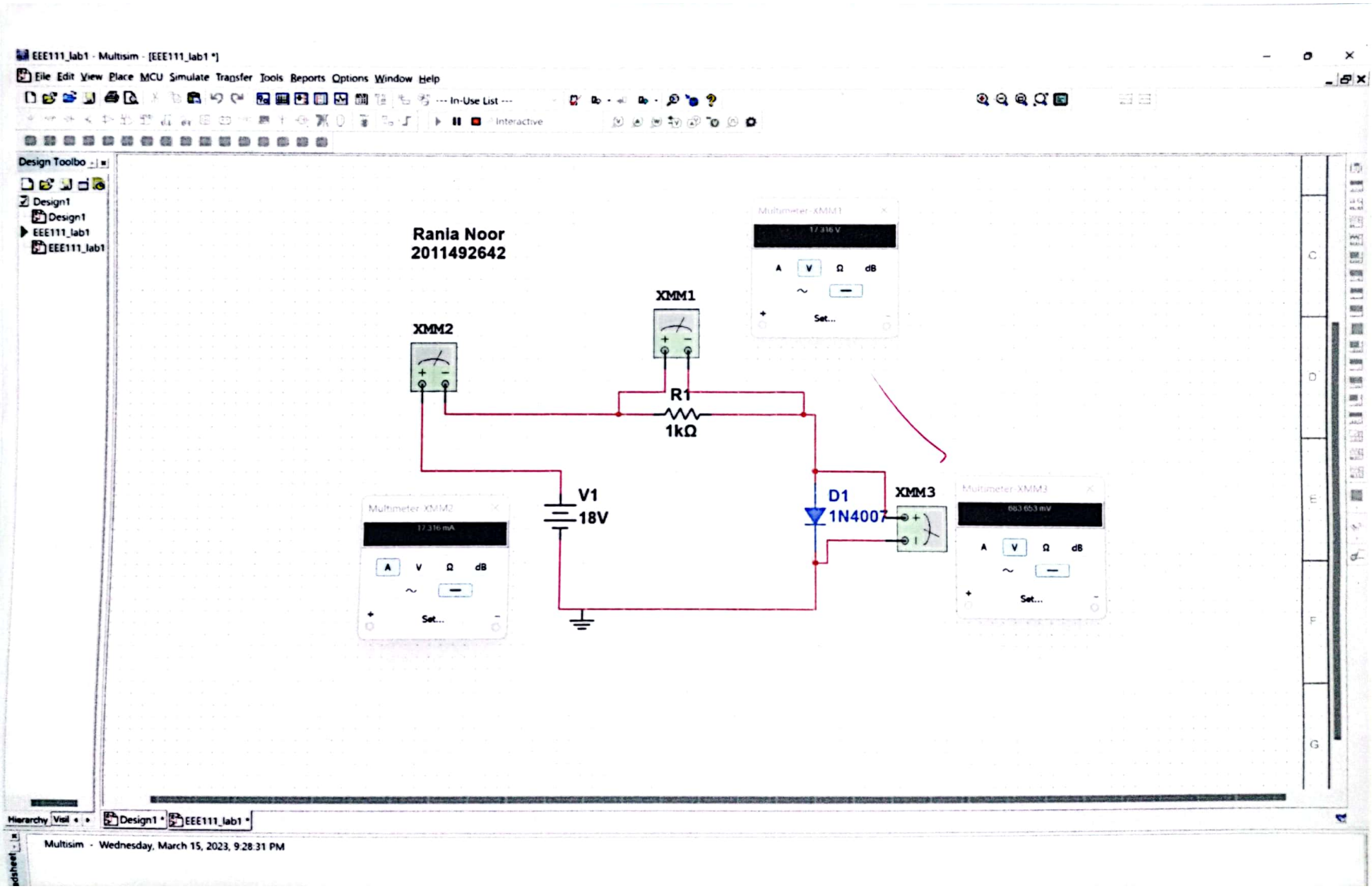
$$\% \text{ Error} = \frac{|\text{Theoritical value} - \text{Measured value}|}{\text{Theoritical value}} \times 100\%.$$

$$= \frac{|3.401 - 3.50|}{3.401} \times 100\%.$$

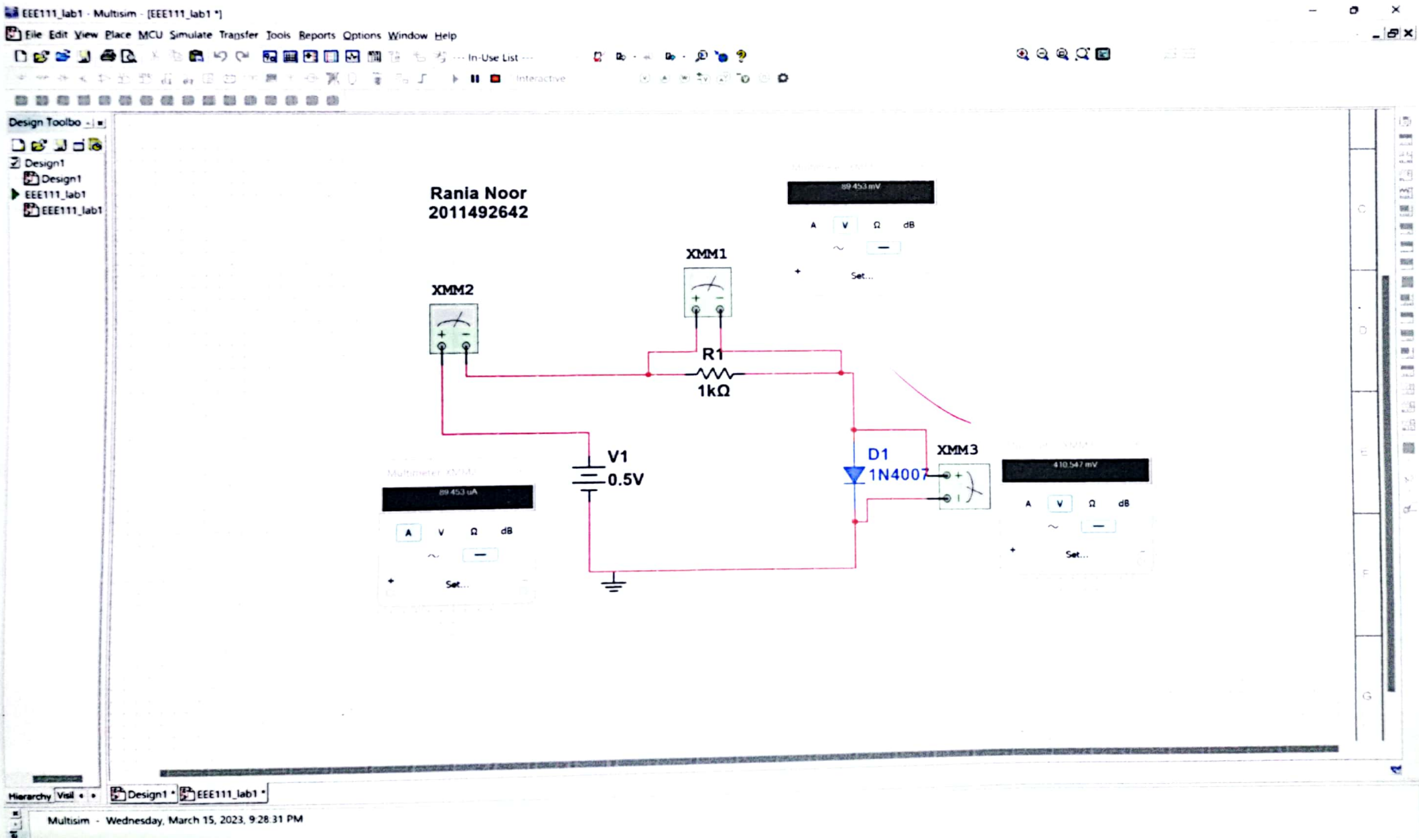
$$= 2.82\%.$$

The percentage error for $V_{dc} = 4V$ with respect to I_d is 2.82%.

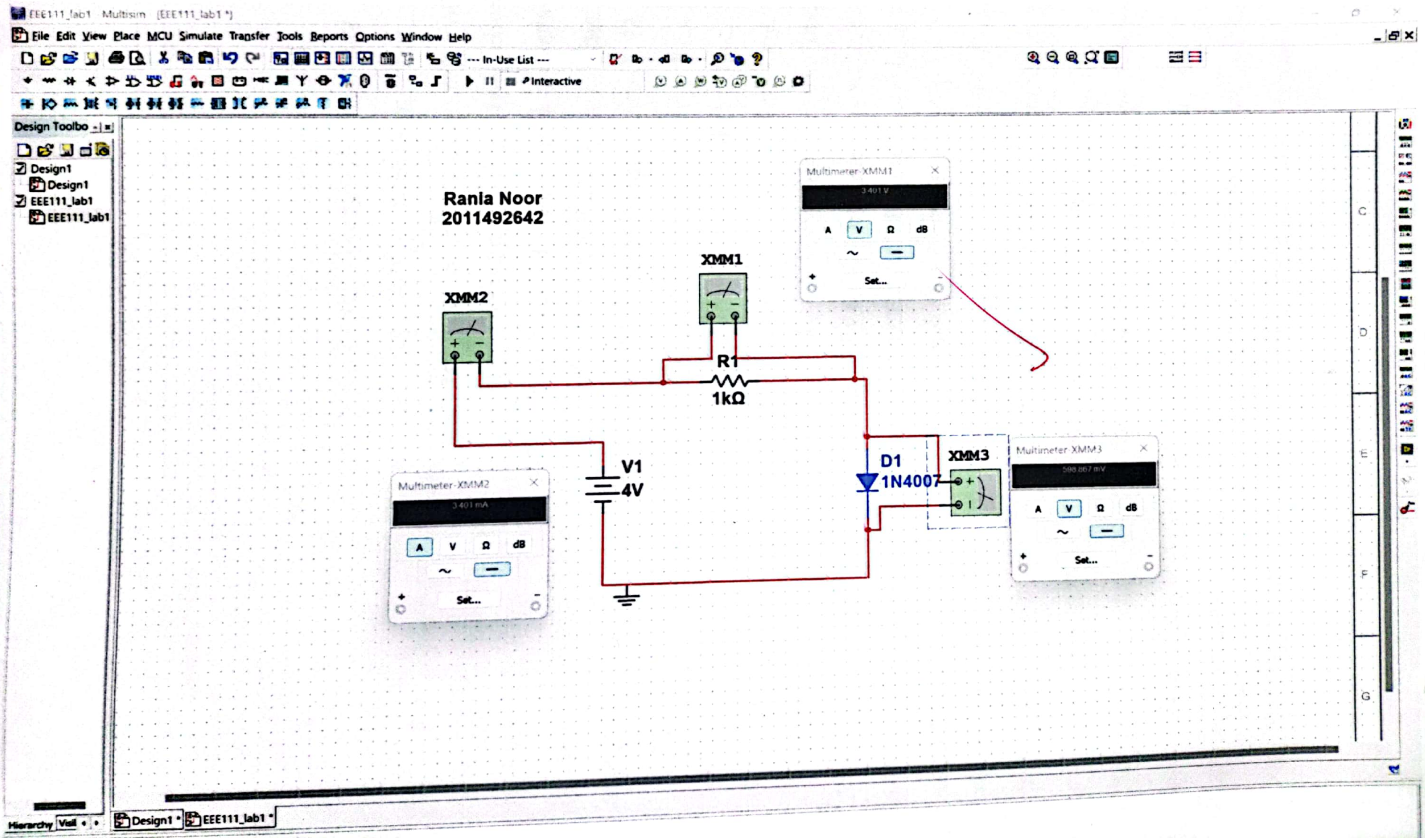
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89.453 uA
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Multimeter-XMM2
Multimeter-XMM3
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Name : farhana Islam
ID - 2021541642

Discussion :

In the lab experiment, we have learned the I-V characteristics of a diode by using measuring the current flowing through it as a function of the voltage applied. our results showed that the diode behaves differently when forward biased and reverse biased. when the diode was forward-biased, the current increased rapidly with voltage until it reached the diode's forward voltage drop. This behaviour is due the p-n junction within the diode, when the diode was reverse biased, the current was very small and independent of the voltage applied, until the reverse breakdown voltage was reached. Our experimental results closely matched the theoritical values for the I-V characteristics of a diode. we also observed that the slope of the I-V curve varied depending on the voltage and current range, which actually related to the diode's dynamic resistance we also learned the practical applications of diodes which includes voltage, rectification etc we also came to know that diodes are commonly

used in electronic circuits due to their simple structure. By doing this lab experiment its provide us a valuable opportunity to explore the uses of diodes and its practical application in electronic circuits. By understanding the I-V characteristics of diode, we can analyze and design the circuits more efficiently.

If we give 6v as input voltage (V_{dc}) we get $I_d = 5.44$ mA (measured); $I_d = 5.377$ mA (theoretical)

So, the percentage of error $= \left| \frac{\text{Measured} - \text{theoretical value}}{\text{theoretical value}} \right|$

$= \frac{5.44\text{ mA} - 5.377\text{ mA}}{5.377\text{ mA}} \times 100\%$

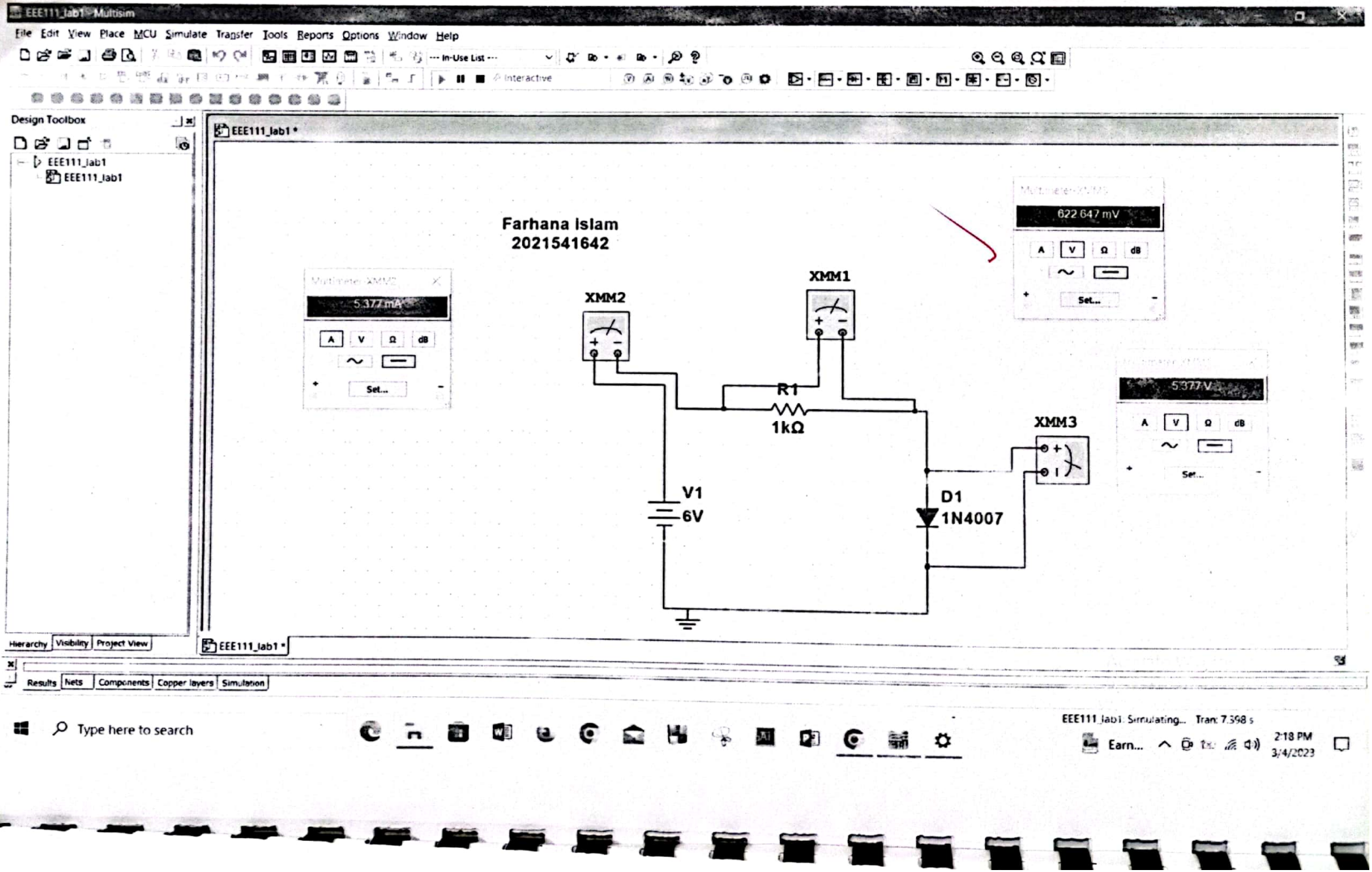
$= 1.17\%$.

In the experiment, we got percentage of error of 1.17%. because of wire, human error and other instrumental error etc.

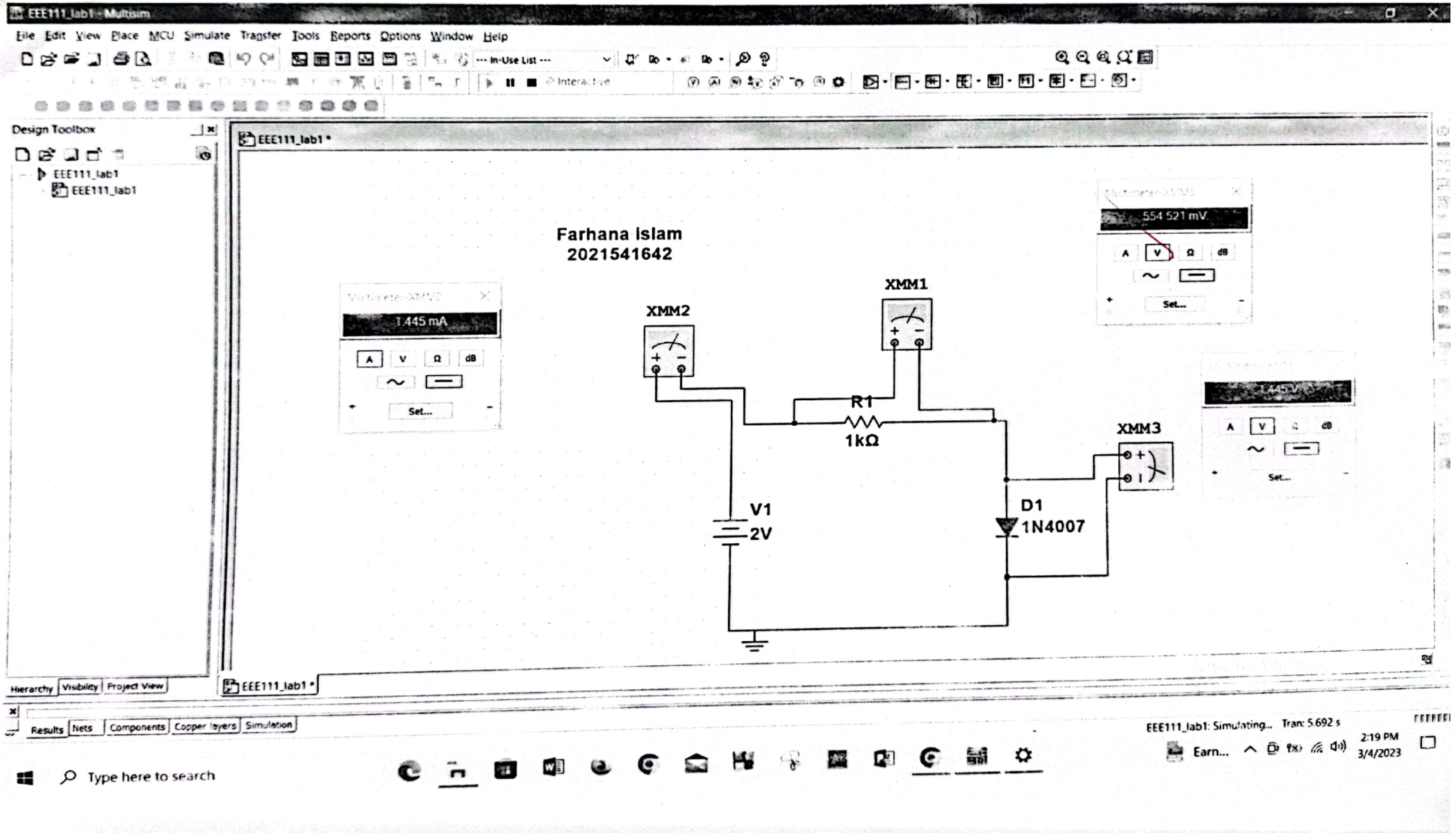
Farhana Islam
2021541642

V_{dc} (volt)	V_d (volt)	V_R (volt)	$I_d = \frac{V_R}{R}$ (mA)
0.1	0.099 V	0 V	0. mA
0.3	0.291 V	0.008 V	0.008 mA
0.5	0.410 V	0.089 V	0.238 mA
0.7	0.461 V	0.236 V	0.410 mA
0.9	0.489 V	0.410 V	0.500 mA
1	0.499 V	0.500 V	1.445 mA
2	0.554 V	1.445 V	2.419 mA
3	0.581 V	2.419 V	3.401 mA
4	0.598 V	3.401 V	5.377 mA
6	0.622 V	5.377 V	7.361 mA
8	0.638 V	7.361 V	9.349 mA
10	0.651 V	9.349 V	11.339 mA
12	0.661 V	11.339 V	13.330 mA
14	0.669 V	13.330 V	15.323 mA
16	0.677 V	15.323 V	17.316 mA
18	0.683 V	17.316 V	17.316 mA
20	0.689 V	19.311 V	19.311 mA

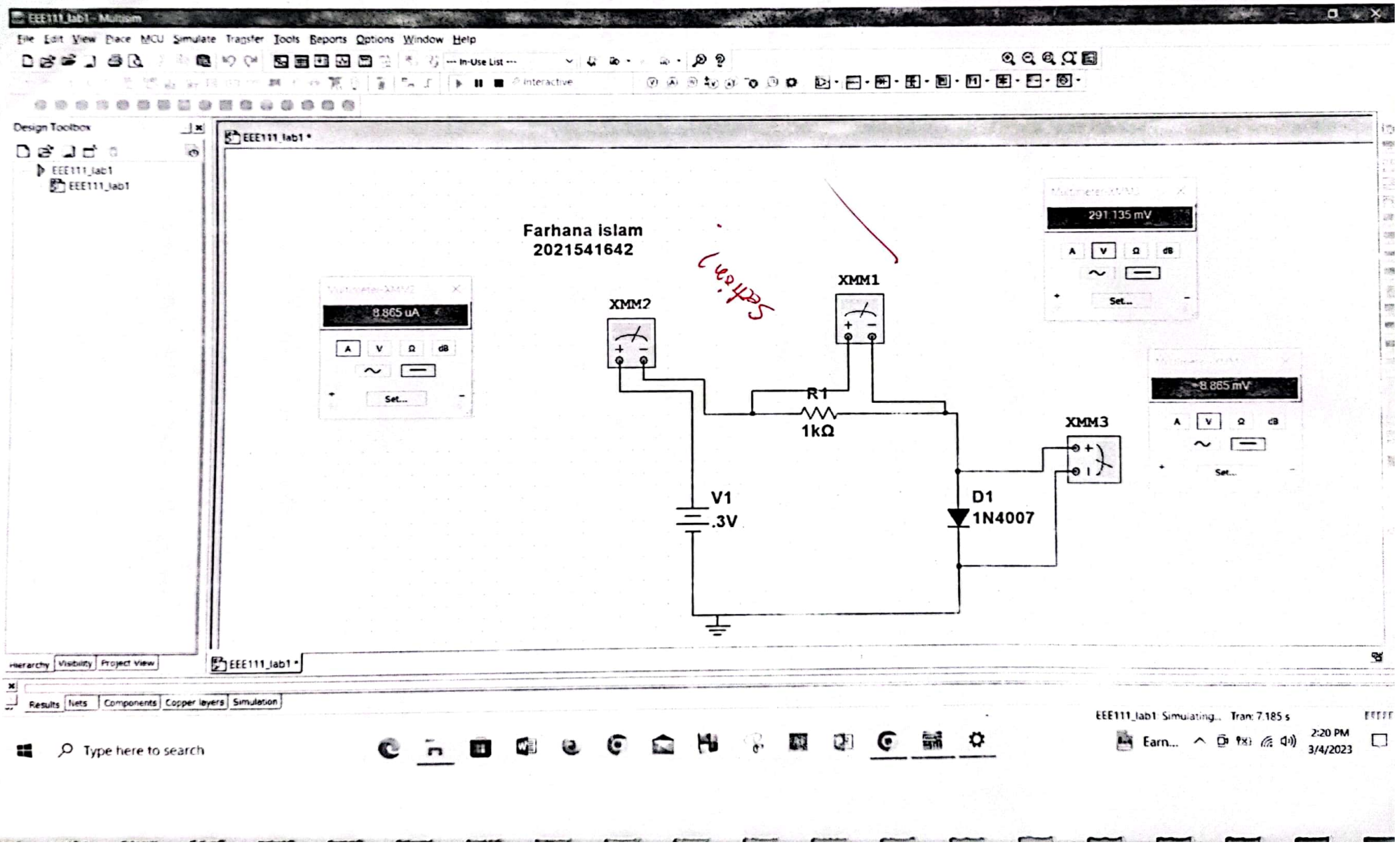
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EEE111_lab1 - Multisim
File Edit View Place MCU Simulate Transfer Tools Reports Options Window Help
Design Toolbox
EEE111_lab1
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Farhana Islam
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622.647 mV
5.377 mA
5.377 V
XMM1
XMM2
XMM3
R1
1kΩ
V1
6V
D1
1N4007
Set...
Hierarchy
Visibility
Project View
Results
Nets
Components
Copper layers
Simulation
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EEE111_lab1 - Multisim
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Farhana Islam
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XMM1
XMM2
XMM3
R1
1kΩ
V1
2V
D1
1N4007
1.445 mA
554.521 mV
Set...
Hierarchy Visibility Project View
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EEE111_lab1 - Multisim
File Edit View Place MCU Simulate Transfer Tools Reports Options Window Help
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XMM2
8.865 uA
XMM1
291.135 mV
R1
1kΩ
XMM3
8.865 mV
V1
.3V
D1
1N4007
Hierarchy
Visibility
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EEE111_lab1 *
Results
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Copper layers
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Dissecussion;

Tanvira Tanrim Phogga (2013914642)

In this experiment we have seen that the theoritical value and measured value is not exactly the sume. It is slithlghy different.

For the experimental value we get the first two values of V_R and I_d are 0. But for V_{de} is 0.1 V_d is zero. Also I_d of measured value slightly flatuats. wheres the theoritical value of V_d and I_d is always the same. The reason of flatuation is the measured value of R is not exactly 1k wheares theoritical value is also. In this circuit we used 6,12V and got 622.642, mV 681.992 mV which is 6.22V and 6.81V But an experimental value it is 0.656V and 0.691V. Due to loss, little bit difference of Resistance the value of V_d in experimental Athanitical will not be same as simulation value.

Since the measured and theoritical value is not same so it is evident that this experiment has some errors.

$$\% \text{ Error} = \frac{|\text{theoritical value} - \text{measured value}|}{\text{theoritical value}} \times 100\%.$$

$$= \left| \frac{9.349 - 9.46}{9.349} \right| \times 100\%.$$

$$= 0.01187 \times 100\%.$$

$$= 1.187 \%$$

6. Result table:

Theoritical value : $R = 1\,k\Omega$

Measured value : $R = 0.987\,k\Omega$.

Vde (volt)	Vd (volt)	V_R (Volt)	$I_d = V_R/R$ (mA)
0.1	0.099 V	0V	0 mA
0.3	0.291 V	0.008 V	0.008 mA
0.7	[illegible]	0.238 V	0.238 mA
0.9	0.461 V	0.410 V	0.410 mA
1	0.489 V	0.500 V	0.500 mA
2	0.499 V	1.445 V	1.445 mA
3	0.554 V	2.419 V	2.419 mA
4	0.581 V	3.401 V	3.401 mA
6	0.598 V	5.377 V	5.377 mA
8	0.622 V	7.361 V	7.361 mA
10	0.638 V	9.349 V	9.349 mA
12	0.651 V	11.339 V	11.339 mA
14	0.669 V	13.330 V	13.330 mA
16	0.677 V	15.323 V	15.323 mA
18	0.683 V	17.316 V	17.316 mA
20	0.689 V	19.311 V	19.311 mA
0.5	0.410 V	0.089 V	0.089 mA

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L-1
C-1
55-1

CamScanner

L-1
C-1
SS-2

